

**ANSWER SCHEME FOR FSPC0034  
CHEMISTRY II TEST 1**

**FOUNDATION SPACE**

**SEMESTER 3 YEAR 2020/2021**

# QUESTION 1

| Question |    | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)      | i  | Endothermic, $\Delta H = +$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|          | ii | Exothermic, $\Delta H = -$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| (b)      | i  | $\text{C}_9\text{H}_{20} (\text{l}) + 14 \text{O}_2 (\text{g}) \rightarrow 9 \text{CO}_2 (\text{g}) + 10 \text{H}_2\text{O} (\text{g})$                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|          | ii | <div style="border: 1px solid black; padding: 10px; margin: 10px 0;"> <math display="block">\Delta H_{\text{Reaction}} = \sum \Delta H_f^{\circ} (\text{Products}) - \sum \Delta H_f^{\circ} (\text{Reactants})</math> </div> $\Delta H^{\circ}_{\text{C}} = [10 \times \Delta H^{\circ}_f (\text{H}_2\text{O}) + 9 \times \Delta H^{\circ}_f (\text{CO}_2)] - [\Delta H^{\circ}_f (\text{C}_9\text{H}_{20}) + 14 \times \Delta H^{\circ}_f (\text{O}_2)]$ $\Delta H^{\circ}_{\text{C}} = [10 \times (-285.8) + 9 \times (-393)] - [-255.2 + 0]$ $\Delta H^{\circ}_{\text{C}} = -6\,139.8 \text{ kJ/mol}$ |

# QUESTION 1

(c)

$$\text{mol} = 4.62 \text{ g} / 114 \text{ g/mol}$$

$$\text{mol} = 0.041 \text{ mol}$$

$$\Delta H = -q / \text{mol}$$

$$q = -(\Delta H \times \text{mol})$$

$$q = -(-5461 \text{ kJ/mol} \times 0.041 \text{ mol})$$

$$q = 221.314 \text{ kJ}$$

$$q = C \cdot \Delta T$$

$$221.314 \text{ kJ} = 7.63 \text{ kJ/}^\circ\text{C} \times \Delta T$$

$$\Delta T = 221.314 \text{ kJ} / 7.63 \text{ kJ/}^\circ\text{C}$$

$$\Delta T = 29.01 \text{ }^\circ\text{C} \text{ (answer range } 28 \text{ }^\circ\text{C} < \text{answer} < 30 \text{ }^\circ\text{C)}$$

# QUESTION 2

| Question |    | Scheme                                                                                                                                                                                                                                                                      |
|----------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)      | i  | $\text{Rate, } R = 3.5 \times 10^{-5} \text{ M s}^{-1} = + 1/2 \text{ d}[\text{H}_2\text{O}]/\text{dT}$<br>Formation rate of water vapour, $\text{H}_2\text{O} = 2 \times R$<br>$= 2 \times 3.5 \times 10^{-5} \text{ M s}^{-1}$<br>$= 7.0 \times 10^{-5} \text{ M s}^{-1}$ |
|          | ii | $\text{Rate} = (\text{change in concentration}) / \text{time taken}$<br>$\text{Time taken} = (\text{change in concentration}) / \text{rate}$<br>$= (0.1 - 0.065) / 3.5 \times 10^{-5} \text{ M s}^{-1}$<br>$= 0.035 / 0.000035 = 1000 \text{ s}$                            |

# QUESTION 2

|     |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (b) | i  | $\frac{k(0.25)^x(0.45)^y}{k(0.25)^x(0.15)^y} = \frac{1.5 \times 10^{-4}}{5.0 \times 10^{-5}}$ $3^y = 3$ <p>y = 1; O<sub>2</sub> is first order</p> $\frac{k(0.50)^x(0.15)^y}{k(0.25)^x(0.15)^y} = \frac{2.0 \times 10^{-4}}{5.0 \times 10^{-5}}$ $2^x = 4$ <p>x = 2; NH<sub>3</sub> is second order</p> <p>Overall order:</p> <p>1 + 2 = 3; third order overall reaction</p>                                                                                |
|     | ii | <p>Rate = k[NH<sub>3</sub>]<sup>2</sup>[O<sub>2</sub>]</p> $k = \text{rate} / ([\text{NH}_3]^2[\text{O}_2]) = 5.0 \times 10^{-5} / [(0.25)^2 \times 0.15]$ $= 5.33 \times 10^{-3} \text{ M}^{-2} \text{ s}^{-1}$ $\text{Rate} = (5.33 \times 10^{-3} \text{ M}^{-2} \text{ s}^{-1}) \times (0.1)^2 \times 0.35$ $= 1.87 \times 10^{-5} \text{ M s}^{-1}$ <p>(Answer range <math>1.80 \times 10^{-5} &lt; \text{answer} &lt; 1.90 \times 10^{-5}</math>)</p> |

## QUESTION 2

|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (c) | i.  | <p>Using equation : <math>\frac{1}{[A]_t} - \frac{1}{[A]_o} = kt</math></p> <p><math>[\text{NO}_2]_t = ?</math>,</p> <p><math>[\text{NO}_2]_o = 0.025 \text{ M}</math>, <math>k = 0.2 \text{ M}^{-1} \text{ s}^{-1}</math>, <math>t = 60 \times 60 = 3600 \text{ s}</math></p> <p><math>1/[\text{NO}_2]_t = kt + 1/[\text{NO}_2]_o = (0.2 \times 3600) + 1/0.025 = 760</math></p> <p><math>[\text{NO}_2]_t = 1/760 = 1.32 \times 10^{-3} \text{ M}</math> or <math>0.00132 \text{ M}</math></p> |
|     | ii. | <p>New concentration = <math>0.005 \text{ M}</math></p> <p>Time taken = <math>(1/[\text{NO}_2]_t - 1/[\text{NO}_2]_o) / k</math></p> <p><math>= (1/0.005 - 1/0.025) / 0.2 = 800 \text{ s}</math></p>                                                                                                                                                                                                                                                                                            |

# QUESTION 3

| Question |     | Scheme                                                                                                                                                                                                                                                              |
|----------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a)       |     | <p>When the rates of the forward and the reverse reactions are equal, equilibrium has been established.</p> <p>For example, when a liquid is placed in a closed container, equilibrium is established when the liquid vaporizes and condenses at the same rate.</p> |
| b)       | i.  | A decrease in pressure decreases $[\text{CH}_3\text{OH}]$ .                                                                                                                                                                                                         |
|          | ii. | An addition of $\text{CO(g)}$ increases $[\text{CH}_3\text{OH}]$ .                                                                                                                                                                                                  |

## QUESTION 3

c)

i.

$$K = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(0.003)^2}{(0.09)(0.1)^3} = \frac{9 \times 10^{-6}}{(9 \times 10^{-2})(1 \times 10^{-3})} = 0.1$$

ii.

The reaction progress is indicated by Q at 3.0 M

$$Q = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(3)^2}{(3)(3)^3} = \frac{1}{(9)} = 0.11$$

iii.

Since  $Q > K$ , the reaction will shift to the left.



## QUESTION 3

|    |     |                                                                                                      |
|----|-----|------------------------------------------------------------------------------------------------------|
| d) | i.  | $\text{AgCrO}_4(\text{s}) \rightleftharpoons 2\text{Ag}^+(\text{aq}) + \text{CrO}_4^{2-}(\text{aq})$ |
|    | ii. | The equilibrium will <u>shift to the left</u> and the solubility of silver chromate decreases.       |